

Continuum Mechanics

Class # 13

09/29/2026

Principal Stretches

Definition: Let $\vec{\phi}: B \rightarrow B$ be an admissible deformation and $F = \nabla \vec{\phi}$ its deformation gradient. Suppose

$$F = RU = VR$$

be the right and left polar decompositions with

$$U = \sqrt{F^T F} \quad V = \sqrt{F F^T}$$

being SPD and R being a rotation, i.e.

$$RR^T = I \quad \det(R) = 1.$$

Let $\{\lambda_i, \vec{u}_i\}$ $\{\lambda_i, \vec{v}_i\}$ be the eigenpairs of U and V , respectively, where

$$\lambda_1 \geq \lambda_2 \geq \lambda_3 > 0$$

and

$$\vec{u}_i \cdot \vec{u}_j = \delta_{ij} \quad \vec{v}_i \cdot \vec{v}_j = \delta_{ij}.$$

We call λ_i the principal stretches associated with F .

$\vec{u}_i$ and $\vec{v}_i$ are called, respectively, the right and left principal directions.

U and V are called the right and left stretch tensors.

Cauchy-Green Strain Tensor

Definition: Consider a general (admissible) deformation

$$\vec{\phi}: B \rightarrow B'$$

with associated deformation gradient $F = \nabla \phi$. The tensor-valued field $C: B \rightarrow \mathcal{V}^2$ defined by

$$C = F^T F$$

is called the right Cauchy-Green strain tensor field associated with ϕ . //

Remark: Observe that C is SPD at each point $\vec{X} \in B$. //

Remark: $C: B \rightarrow \mathcal{V}^2$ contains information about stretches only. Indeed, let

$$F = RU$$

be the right polar decomposition of F , where R is a rotation matrix, i.e.,

$$RR^T = I \quad \det(R) = 1,$$

and $U = \sqrt{F^T F}$ is SPD. Then

$$C = F^T F = U^2.$$

This fact makes C a "better" measure of strain.
See Exercise 14. //

Proposition Suppose that $U \in V^2$ is symmetric and positive definite (SPD). Then the eigenvalues are real and positive:

$$\lambda_1 \geq \lambda_2 \geq \lambda_3 > 0$$

Furthermore there is an orthonormal basis of eigenvectors $\{\vec{u}_1, \vec{u}_2, \vec{u}_3\}$, that is,

$$U \vec{u}_i = \lambda_i \vec{u}_i, \quad (\text{no sum}) \quad 1 \leq i \leq 3,$$

and

$$\vec{u}_i \cdot \vec{u}_j = \delta_{ij}, \quad 1 \leq i, j \leq 3.$$

Furthermore,

$$U = \sum_{i=1}^3 \lambda_i \vec{u}_i \otimes \vec{u}_i.$$

The matrix U^2 is also SPD. Its eigenpairs are precisely

$$\{\lambda_i^2, \vec{u}_i\}, \quad 1 \leq i \leq 3.$$

Therefore,

$$U^2 = \sum_{i=1}^3 \lambda_i^2 \vec{u}_i \otimes \vec{u}_i.$$

Proof: Exercise.

Result (4.4): Suppose $\vec{X} \in B$ is arbitrary, and Ω is the ball of radius α centered at $\vec{X}$. Let $\vec{e}, \vec{d} \in \mathcal{N}$ be any unit vectors and set

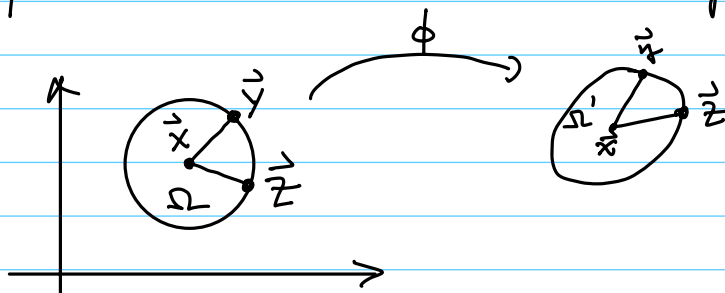
$$\vec{Y} := \vec{X} + \alpha \vec{e} \quad \Rightarrow \quad |\vec{Y} - \vec{X}| = \alpha$$

$$\vec{Z} := \vec{X} + \alpha \vec{d} \quad \Rightarrow \quad |\vec{Z} - \vec{X}| = \alpha$$

Set

$$\vec{x} = \phi(\vec{X}), \quad \vec{y} := \phi(\vec{Y}), \quad \vec{z} := \phi(\vec{Z}),$$

where $\phi: B \rightarrow B'$ is an admissible deformation.



Define

$$\lambda(\vec{e}) = \sqrt{\vec{e} \cdot C \vec{e}}$$

$$\cos(\Theta(\vec{e}, \vec{d})) = \frac{\vec{e} \cdot C \vec{d}}{\sqrt{\vec{e} \cdot C \vec{e}} \sqrt{\vec{d} \cdot C \vec{d}}} \quad \Theta(\vec{e}, \vec{d}) \in [0, \pi]$$

Let $\mathcal{S} \in [0, \pi]$ denote the angle between $\vec{v} = \vec{y} - \vec{x}$ and $\vec{w} = \vec{z} - \vec{x}$. Then

$$(4.5)_1 \quad \frac{|\vec{y} - \vec{x}|}{|\vec{Y} - \vec{X}|} \xrightarrow{\alpha \downarrow 0} \lambda(\vec{e})$$

$$(4.5)_2 \quad \frac{|\vec{z} - \vec{x}|}{|\vec{Z} - \vec{X}|} \xrightarrow{\alpha \downarrow 0} \lambda(\vec{d})$$

$$(4.6) \quad \cos(\mathcal{S}) \xrightarrow{\alpha \downarrow 0} \cos(\Theta(\vec{e}, \vec{d}))$$

Proof: Pages 122-123. ///

Remark: $\lambda(\vec{e})$ is the limiting value of the ratio of

$$\frac{|\vec{F}(\vec{Y}) - \vec{F}(\vec{X})|}{|\vec{Y} - \vec{X}|} \equiv \frac{\text{deformed length}}{\text{initial length}}$$

as $\vec{Y}$ tends to $\vec{X}$. For this reason $\lambda(\vec{e})$ is called the stretch in the direction $\vec{e}$ at $\vec{X}$. //

Proposition: Let $\vec{u}_i$ be a right principal direction at $\vec{X}$ corresponding to the principal stretch λ_i , i.e.,

$$U\vec{u}_i = \lambda_i \vec{u}_i \quad (\text{no sum})$$

where

$$F = RU.$$

Then

$$\lambda(\vec{u}_i) = \lambda_i$$

Proof: By definition

$$\lambda(\vec{u}_i) = \sqrt{\vec{u}_i C \vec{u}_i}$$

But, recall that

$$C = U^2$$

and

$$C\vec{u}_i = \lambda_i^2 \vec{u}_i, \quad \lambda_i > 0. \quad (\text{no sum})$$

Hence

$$\lambda(\vec{u}_i) = \sqrt{\vec{u}_i \cdot (\lambda_i^2 \vec{u}_i)}$$

$$= \lambda_i //$$

Proposition: let $\phi: B \rightarrow B'$ be an admissible deformation with deformation gradient $F = \nabla \vec{\phi}$ and right Cauchy - Green strain tensor

$$C = F^T F = U^2.$$

Suppose that the principal stretches are ordered so that

$$\lambda_1 \geq \lambda_2 \geq \lambda_3 > 0.$$

Then

$$\max_{\vec{e} \in N} \lambda(\vec{e}) = \lambda_1 = \lambda(\vec{u}_1)$$

where $\{\vec{u}_1, \vec{u}_2, \vec{u}_3\}$ are the associated principal directions.

Proof: let $\vec{e} \in N$ ($N \equiv$ the set of unit vectors in V) be arbitrary. Then there scalars $c_i \in \mathbb{R}$ such that

$$\vec{e} = \sum_{i=1}^3 c_i \vec{u}_i$$

with

$$C \vec{u}_i = \lambda_i^2 \vec{u}_i \quad (\text{no sum})$$

We have

$$\lambda^2(\vec{e}) = \vec{e} \cdot (C \vec{e})$$

$$= \left(\sum_{j=1}^3 c_j \vec{u}_j \right) \cdot \left(C \left(\sum_{i=1}^3 c_i \vec{u}_i \right) \right)$$

$$= \left(\sum_{j=1}^3 c_j \vec{u}_j \right) \cdot \left(\sum_{i=1}^3 \lambda_i^2 c_i \vec{u}_i \right)$$

$$= \sum_{j=1}^3 c_j \sum_{i=1}^3 \lambda_i^2 c_i (\vec{u}_j \cdot \vec{u}_i)$$

$$= \sum_{j=1}^3 c_j \sum_{i=1}^3 \lambda_i^2 c_i \delta_{ij}$$

$$= \sum_{i=1}^3 \lambda_i^2 c_i^2$$

$$\leq \sum_{i=1}^3 \lambda_1^2 c_i^2$$

$$= \lambda_1^2 \sum_{i=1}^3 c_i^2$$

$$= \lambda_1^2$$

Thus

$$\lambda(\vec{e}) \leq \lambda_1 \quad \forall \vec{e} \in \mathcal{N}.$$

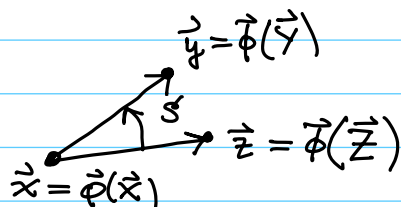
But

$$\lambda(\vec{u}_1) = \lambda_1$$

Thus,

$$\max_{\vec{e} \in \mathcal{N}} \lambda(\vec{e}) = \lambda_1 \quad //$$

Remark: The angle $\theta(\vec{e}, \vec{d})$ is the limiting value of $s \in [0, \pi]$



as $s \searrow 0$ or as $\vec{y}$ and $\vec{z}$ tend to $\vec{x}$, along directions $\vec{e}$ and $\vec{d}$.

let $\Theta(\vec{e}, \vec{d}) \in [0, \pi]$ denote the angle between $\vec{e}$ and $\vec{d}$, the quantity

$$\gamma(\vec{e}, \vec{d}) = \Theta(\vec{e}, \vec{d}) - \theta(\vec{e}, \vec{d})$$

is called the shear between directions $\vec{e}$ and $\vec{d}$ at $\vec{x}$.

Proposition: The shear between any two right principal directions (orthonormal eigenvectors of $\mathbf{U}$) is zero.

Proof Exercises 18 and 19. ///

Result (4.5): Let C_{ij} be the components of the Cauchy-Green strain tensor $\mathbf{C}: \mathcal{B} \rightarrow \mathcal{V}^2$ in a coordinate frame $\mathcal{F} = \{\vec{e}_i\}$. Then, for any point $\vec{x} \in \mathcal{B}$ we have

$$C_{ii} = \lambda^2(e_i)$$

$$C_{ij} = \lambda(\vec{e}_i) \lambda(\vec{e}_j) \sin(\gamma(\vec{e}_i, \vec{e}_j)) \quad (\text{no sum})$$

Proof: Recall that

$$C_{ij} = \vec{e}_i \cdot (\mathbf{C} \vec{e}_j)$$

By definition,

$$\cos(\Theta(\vec{e}_i, \vec{e}_j)) = \frac{\vec{e}_i \cdot (\mathbf{C} \vec{e}_j)}{\sqrt{\vec{e}_i \cdot (\mathbf{C} \vec{e}_i)} \sqrt{\vec{e}_j \cdot (\mathbf{C} \vec{e}_j)}} \quad (\text{no sum})$$

Thus

$$C_{ij} = \lambda(\vec{e}_i) \lambda(\vec{e}_j) \cos(\Theta(\vec{e}_i, \vec{e}_j)) \quad (\text{no sum})$$

Also, for $i \neq j$,

$$\begin{aligned}\gamma(\vec{e}_i, \vec{e}_j) &= \Theta(\vec{e}_i, \vec{e}_j) - \theta(\vec{e}_i, \vec{e}_j) \\ &= \frac{\pi}{2} - \theta(\vec{e}_i, \vec{e}_j)\end{aligned}$$

From above,

$$\begin{aligned}C_{ij} &= \lambda(\vec{e}_i) \lambda(\vec{e}_j) \cos\left(\frac{\pi}{2} - \gamma(\vec{e}_i, \vec{e}_j)\right) \\ &= \lambda(\vec{e}_i) \lambda(\vec{e}_j) \sin(\gamma(\vec{e}_i, \vec{e}_j))\end{aligned}$$

For $i = j$

$$\begin{aligned}C_{ii} &= \vec{e}_i \cdot (C \vec{e}_i) \\ &= \lambda^2(\vec{e}_i) \quad \text{///}\end{aligned}$$

Rigid Deformations

Definition: A homogeneous deformation $\vec{\phi}: B \rightarrow B'$ is called rigid iff

$$\vec{\phi}(\vec{X}) = \vec{c} + Q \vec{X}$$

where $\vec{c} \in V$ and $Q \in V^2$ is a rotation, i.e.

$$QQ^T = I \text{ and } \det(Q) = 1.$$

Proposition: For rigid deformations, the right Cauchy Green strain tensor is the identity. This type of deformation produces no strain.

Proof: $F = Q$ for this homogeneous deformation.

$$C(\vec{X}) = F^T(\vec{X}) F(\vec{X}) = Q^T Q = I_q$$

for all $\vec{X} \in B$. //

Remark: For such a deformation, the relative position and orientation between any three points in B is unchanged by $\vec{\phi}$.

Infinitesimal Strain

Definition: Consider an admissible deformation $\phi: B \rightarrow B'$ with associated displacement field $\vec{u}: B \rightarrow V$:

$$\vec{u}(\vec{X}) = \vec{\phi}(\vec{X}) - \vec{X}$$

Define $E: X \rightarrow V^2$ by

$$E = \text{sym}(\nabla \vec{u}) = \frac{1}{2} (\nabla \vec{u} + (\nabla \vec{u})^T)$$

E is called the infinitesimal Strain tensor field associated with $\vec{\phi}$.

Proposition: We have

$$E = \frac{1}{2}(C - I) - \frac{1}{2}(\nabla \vec{u})^T \nabla \vec{u},$$

where C is the right Cauchy-Green strain tensor associated to $\vec{\phi}$.

Proof: By definition,

$$\nabla \vec{u} = \nabla \vec{\phi} - I$$

$$= F - I$$

Thus,

$$E = \text{sym}(F - I) = \frac{1}{2}(F - I) + \frac{1}{2}(F^T - I).$$

On the other hand,

$$\frac{1}{2}(C - I) - \frac{1}{2}(\nabla \vec{u})^T \nabla \vec{u}$$

$$= \frac{1}{2}(F^T F - I) - \frac{1}{2}(F^T - I)(F - I)$$

$$= \cancel{\frac{1}{2}F^T F} - \frac{1}{2}I - \cancel{\frac{1}{2}F^T F} + \frac{1}{2}F + \frac{1}{2}F^T - \frac{1}{2}I$$

$$= \text{sym}(F - I)$$

$$= E \quad ///$$

Remark: E is useful in the cases for which

$$|\nabla \vec{u}| \ll 1,$$

Say

$$|\nabla \vec{u}| = \mathcal{O}(\varepsilon).$$

Then

$$E = \frac{1}{2}(C - I) + O(\varepsilon^2).$$

When displacements are small,

$$E \approx \frac{1}{2}(C - I).$$

This makes E a useful convenient measure of strain, as we shall see. //

Results (4.6): Let E_{ij} be the components of E in the frame $\mathcal{F} = \{\vec{e}_i\}$. If terms of order $O(\varepsilon^2)$ are sufficiently small as to be neglected, then

$$E_{ii} \approx \lambda(\vec{e}_i) - 1 \quad (\text{no sum})$$

and

$$E_{ij} \approx \frac{1}{2} \sin(\gamma(\vec{e}_i, \vec{e}_j)) \quad i \neq j$$

where $\lambda(\vec{e}_i)$ is the stretch in direction $\vec{e}_i$ and $\gamma(\vec{e}_i, \vec{e}_j)$ is the shear between $\vec{e}_i$ and $\vec{e}_j$.

Proof: Pages 126 - 127.

Infinitesimally Rigid Deformations

Definition: A homogeneous deformation $\phi: B \rightarrow B'$ is called infinitesimally rigid iff the displacement field has the form

$$\vec{u}(\vec{X}) = \vec{c} + W\vec{X},$$

for all $\vec{X} \in B$, where $W \in V^2$ is skew symmetric, i.e.,
 $W^T = -W$

Proposition: For an infinitesimally rigid deformation

$$E(\vec{X}) = 0 \quad (\text{zero tensor})$$

for all $\vec{X} \in B$. Thus there is no strain for such a deformation.

Proof: Observe that

$$\nabla \vec{u} = W$$

Thus

$$E = \frac{1}{2}(\nabla \vec{u} + (\nabla \vec{u})^T)$$

$$= \frac{1}{2}(W - W)$$

$$= 0$$

Using Result 4.6

$$0 = E_{ii} \approx \lambda(\vec{e}_i) - 1$$

Thus

$$\lambda(\vec{e}_i) \approx 1 \quad (\text{no stretch})$$

And

$$0 = E_{ij} \approx \frac{1}{2} \Delta(\vec{e}_i, \vec{e}_j)$$

Thus

$$\Delta(\vec{e}_i, \vec{e}_j) \approx 0 \quad (\text{no shear}). \quad ///$$